

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 Total Marks: 40

Annexure A

Code of Conduct:

I BALAJI D (Reg. No: 192512368) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Department: B.E ECE

Signature of the student: BALAJI.D

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu

II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles

III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

(a) Propositional Logic Representation

Let F = Fever, C = Cough, R = Rash, B = Breathlessness, Flu = Possible Flu, Mea = Possible Measles, Pne = Possible Pneumonia.

Rule 1: $F \wedge C \rightarrow \text{Flu}$

Rule 2: $F \wedge R \rightarrow \text{Mea}$

Rule 3: $C \wedge B \rightarrow \text{Pne}$

Facts (Patient A): F = True, C = True, B = True

(b) Forward Chaining

Step 1: Initial facts — F = True, C = True, B = True.

Step 2: Test Rule 1 ($F \wedge C \rightarrow \text{Flu}$). Both F and C are True $\rightarrow$ Rule fires $\rightarrow$ Flu = True derived.

Step 3: Test Rule 2 ($F \wedge R \rightarrow \text{Mea}$). R is unknown/False $\rightarrow$ Rule does not fire.

Step 4: Test Rule 3 ($C \wedge B \rightarrow \text{Pne}$). Both C and B are True $\rightarrow$ Rule fires $\rightarrow$ Pne = True derived.

Answer: Derived diagnoses for Patient A = {Possible Flu, Possible Pneumonia}.

(c) Backward Chaining — Goal: Pne (Patient A has Pneumonia)

Goal reduction tree:

Pne

└ Rule 3: $C \wedge B \rightarrow \text{Pne}$

└ Sub-goal C $\rightarrow$ check facts $\rightarrow$ C = True ✓

└ Sub-goal B $\rightarrow$ check facts $\rightarrow$ B = True ✓

Answer: Both sub-goals are satisfied by known facts, so the goal Pne is proved. Patient A has Pneumonia = True.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]

II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]

III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed(CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

(a) Unification

Unify Vehicle(x) with the fact Vehicle(Ambulance): $\theta_1 = \{x/Ambulance\}$.

Apply θ_1 to EmergencyType(x) $\rightarrow$ EmergencyType(Ambulance), which matches the given fact ✓

Answer: MGU $\vartheta = \{x/Ambulance\}$. Applying ϑ to Rule 1's conclusion gives ClearPath(Ambulance).

(b) Resolution Proof of Proceed(CarA)

CNF form of the rules:

Rule 1: $\neg Vehicle(x) \vee \neg EmergencyType(x) \vee ClearPath(x)$

Rule 2: $\neg ClearPath(x) \vee GreenSignal(x)$ (and $\neg ClearPath(x) \vee \neg RedSignal(x)$)

Rule 3: $\neg Vehicle(x) \vee \neg Behind(x,y) \vee \neg GreenSignal(y) \vee Proceed(x)$

Facts (unit clauses): Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA,Ambulance)

Negated goal: $\neg Proceed(CarA)$

Step 1: Resolve Vehicle(Ambulance) with Rule 1 $\{x/Ambulance\} \rightarrow \neg EmergencyType(Ambulance) \vee ClearPath(Ambulance)$

Step 2: Resolve with EmergencyType(Ambulance) $\rightarrow ClearPath(Ambulance)$

Step 3: Resolve ClearPath(Ambulance) with Rule 2 $\{x/Ambulance\} \rightarrow GreenSignal(Ambulance)$

Step 4: Resolve Vehicle(CarA), Behind(CarA,Ambulance) and GreenSignal(Ambulance) with Rule 3 $\{x/CarA, y/Ambulance\} \rightarrow Proceed(CarA)$

Step 5: Resolve Proceed(CarA) with the negated goal $\neg Proceed(CarA) \rightarrow$ empty clause $\square$

Answer: The empty clause is derived, so Proceed(CarA) is proved true.

(c) Sufficiency for Two Simultaneous Emergency Vehicles

The current KB is not sufficient. If two ambulances satisfy Rule 1 independently, both would be assigned ClearPath and GreenSignal at the same time, which can create a logical conflict when their routes intersect.

Additional rules/facts needed:

- A Conflicts(x,y) predicate to identify vehicles whose routes intersect.
- A Priority(x,y) rule to decide precedence (e.g., based on arrival time, severity, or distance to intersection).
- A mutual-exclusion constraint: $\neg(GreenSignal(x) \wedge GreenSignal(y))$ when Conflicts(x,y) holds, unless one yields.

Answer: Justification: without these additions the KB cannot resolve contradictory GreenSignal assignments, so it needs explicit conflict-detection and priority rules to remain consistent.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

(a) CNF Conversion

Let D = SoilDry, I = IrrigationNeeded, W = CropWheat, A = ApplyDripMethod, R = CropAtRisk.

P1: $D \rightarrow I \Rightarrow$ eliminate implication $\Rightarrow \neg D \vee I$ (already a single clause)

P2: $(I \wedge W) \rightarrow A \Rightarrow \neg(I \wedge W) \vee A \Rightarrow$ move $\neg$ inward (De Morgan) $\Rightarrow \neg I \vee \neg W \vee A$

P3: $(\neg A \wedge W) \rightarrow R \Rightarrow \neg(\neg A \wedge W) \vee R \Rightarrow$ move $\neg$ inward $\Rightarrow (A \vee \neg W) \vee R \Rightarrow A \vee \neg W \vee R$

Facts as unit clauses: D, W

(b) Resolution Refutation for ApplyDripMethod

Clauses: (1) $\neg D \vee I$ (2) $\neg I \vee \neg W \vee A$ (3) $A \vee \neg W \vee R$ (4) D (5) W Negated goal: (6) $\neg A$

Step 1: Resolve (4) D with (1) $\neg D \vee I \rightarrow I$

Step 2: Resolve I with (2) $\neg I \vee \neg W \vee A \rightarrow \neg W \vee A$

Step 3: Resolve $(\neg W \vee A)$ with (5) W $\rightarrow A$

Step 4: Resolve A with (6) $\neg A \rightarrow$ empty clause $\square$

Resolution tree: D & $(\neg D \vee I) \rightarrow I$; I & $(\neg I \vee \neg W \vee A) \rightarrow (\neg W \vee A)$; $(\neg W \vee A)$ & W $\rightarrow A$; A & $\neg A \rightarrow \square$

Answer: The empty clause is derived, so ApplyDripMethod is proved true.

(c) Effect of Removing the Fact 'SoilDry'

Without D = True, Rule P1 ($D \rightarrow I$) can no longer fire, so IrrigationNeeded (I) can no longer be derived.

Without I, Rule P2 ($(I \wedge W) \rightarrow A$) can no longer fire, so ApplyDripMethod (A) can no longer be proved true by the KB.

Under strict (classical/monotonic) resolution: since $\neg A$ is not entailed by the KB either, CropAtRisk (which needs $\neg A \wedge W$) cannot be logically proved — the outcome becomes undetermined.

Under a closed-world / negation-as-failure assumption (common in rule-based expert systems): since ApplyDripMethod cannot be proved, it is treated as False, so $\neg A \wedge W$ becomes True, and Rule P3 fires, giving CropAtRisk = True.

Answer: Conclusion: CropAtRisk does not follow under strict classical entailment, but it does follow under the closed-world assumption typically used by rule-based agents. This illustrates the difference between monotonic logical proof and default (non-monotonic) reasoning.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench[1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze[1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter[x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

(a) Belief State via Modus Ponens

From Stench[1,2] = True and Rule 1 (Modus Ponens) $\rightarrow$ WumpusAdjacent[1,2] = True

From Breeze[1,1] = True and Rule 2 (Modus Ponens) $\rightarrow$ PitAdjacent[1,1] = True

From Glitter[2,2] = True and Rule 3 (Modus Ponens) $\rightarrow$ Gold[2,2] = True

Answer: Derived facts: WumpusAdjacent[1,2], PitAdjacent[1,1], Gold[2,2].

(b) Forward Chaining — Possible Wumpus and Pit Locations

WumpusAdjacent[1,2] means the Wumpus is in one of the cells adjacent to [1,2]: candidate set = {[1,1], [1,3], [2,2]}.

PitAdjacent[1,1] means the Pit is in one of the cells adjacent to [1,1]: candidate set = {[1,2], [2,1]}.

The agent started safely at [1,1] (no stench/breeze reported when it was first entered as the start cell), so the Wumpus is not at [1,1]. No stench is perceived at [2,2] itself, so by the contrapositive of Rule 1 the Wumpus cannot be adjacent to [2,2] either, which rules out [2,2] as the Wumpus's own cell being flagged again — combined with the [1,1] elimination, the remaining candidate is [1,3].

Answer: Most probable Wumpus location: [1,3]. Most probable Pit location: one of {[1,2], [2,1]} — further percepts (e.g., breeze/stench at [1,2] or [2,1]) are needed to fully disambiguate.

(c) Safety Evaluation of Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

[1,1]: Starting cell, already occupied safely by the agent $\rightarrow$ Safe.

[2,2]: No stench or breeze is perceived here (only Glitter), so by Rule 4, $\neg$ Wumpus[2,2] $\wedge$ $\neg$ Pit[2,2] $\rightarrow$ Safe[2,2] = True.

[1,2]: This cell is where Stench is perceived, which only implies the Wumpus is adjacent to [1,2] (not necessarily inside it). Since the Wumpus is most likely at [1,3] (from part b), [1,2] is probably not occupied by the Wumpus. However, [1,2] is also one of the two candidate cells for the Pit (from PitAdjacent[1,1]), so $\neg$ Pit[1,2] cannot yet be confirmed.

Answer: Conclusion: [1,1] and [2,2] can be logically confirmed as Safe, but [1,2] remains a candidate Pit location and cannot yet be proved safe from the given percepts alone. The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ therefore carries residual risk at the [1,2] step; the agent should gather an additional percept (e.g., check for breeze at [2,1]) before committing to that step, or consider an alternative route to [2,2] if one exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand